LAB # 4
Reproduce Basic Device Configuration Sequence to Ensure Device Connectivity in Layer 3 Devices using Cisco Packet Tracer Objectives

Objectives

- To trace configuration procedures for basic layer 3 devices using command line interface in Packet Tracer.
- To explain and show network connectivity using Cisco packet tracer.

Pre-Lab Exercise

Read this experiment in its entirety to become familiar with objectives of this lab. Have Cisco Packet Tracer software program installed on your PC and review how to simulate/designs network topologies within it. You may record the terms and sections that require more elaboration for reference. The instructor may provide the class some time to reflect upon these before proceeding with the lab.

Cisco Packet Tracer

Cisco Packet Tracer is a powerful network simulation program that allows students to experiment with network behavior and ask "what if" questions. As an integral part of the Networking Academy comprehensive learning experience, Packet Tracer provides simulation, visualization, authoring, assessment, and collaboration capabilities and facilitates the teaching and learning of complex technology concepts.

Packet Tracer supplements physical equipment in the classroom by allowing students to create a network with an almost unlimited number of devices, encouraging practice, discovery, and troubleshooting. You will simulate network topologies on Cisco Packet Tracer.

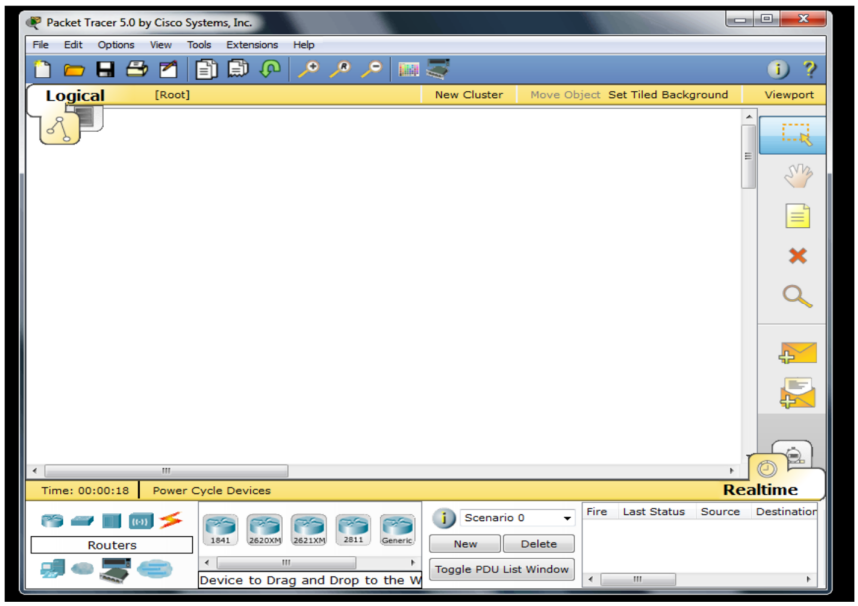

Figure 10: Cisco Packet Tracer

Command Liner Interface (CLI)

After the interface status messages appear and you press Enter, the *Switch>or Router>*prompt will pop up. This is called *user exec mode*, or user mode for short, and although it's mostly used to view statistics. There are three modes each with access to different command sets:

User exec mode—this is the first mode a user has access to after logging into the Switch or Router. This mode allows the user to execute only the basic commands, such as those that show the system's status. The system cannot be configured or restarted from this mode.

Privileged mode—this mode allows users to view the system configuration, restart the system, and enter configuration mode. It also allows all the commands that are available in user mode. You enter it via the enable command like this:

Switch>enable

Switch#

Privileged mode can be identified by the # prompt following the router or switch name. You can go back from privileged mode into user mode by using the **disable** command:

Switch#**disable**

Switch>

Configuration mode— this mode allows users to modify the running system configuration. To enter configuration mode, enter the command **configure terminal** from privileged mode.

Switch#**configure terminal**

Switch(config)#

Configuration mode has various sub modes, starting with global configuration mode, which can be identified by the (config)# prompt following the router name. As the configuration mode submodes change depending on what is being configured, the words inside the parentheses change.

Mode	Definition
User exec mode	Limited to basic monitoring commands
Privileged exec mode	Provides access to all other router commands
Global configuration mode	Commands that affect the entire system
Specific Configuration mode	Commands that affect interfaces/processes only
Setup mode	Interactive configuration dialog

Enhanced Editing Commands

Following commands are very useful while working with CISCO IOS.

Command	Meaning
Ctrl+A	Moves your cursor to the beginning of the line
Ctrl+E	Moves your cursor to the end of the line
Esc+B	Moves back one word
Ctrl+B Ctrl+F	Moves back one character Moves forward one character
Esc+F	Moves forward one word
Ctrl+D/ Backspace	Deletes a single character
Ctrl+R	Redisplays a line

Ctrl+U	Erases a line
Ctrl+W	Erases a word
Ctrl+Z	Ends configuration mode and returns EXEC

In-Lab Exercise

Topology Diagram

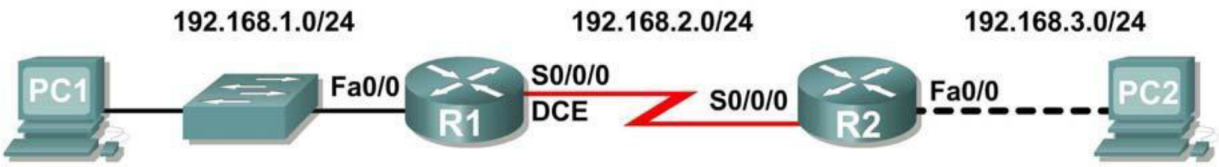

Figure 11: Network Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.1	255.255.255.0	NA
	S0/0/0	192.168.2.1	255.255.255.0	NA
R2	Fa0/0	192.168.3.0	255.255.255.0	NA
	S0/0/0	192.168.2.2	255.255.255.0	NA
PC1	NA	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NA	192.168.3.10	255.255.255.0	192.168.3.1

Task 1: Cable the Ethernet and Serial Links of the Network

Step 1: Answer the following questions.

What type of cable is used to connect the Ethernet interface on a host PC to the Ethernet interface on a switch?

straight through

What type of cable is used to connect the Ethernet interface on a switch to the Ethernet interface on a router?

straight through

What type of cable is used to connect the Ethernet interface on a router to the Ethernet interface on a host PC?

crossover

What type of cable is used to connect the Serial interface on a router R1 to the Serial interface on a router R2?

serial dce cable

Task 2: Perform basic IOS command line interface operations

Step 1: Establish a terminal session to router R1.

The console port is a management port used to provide out-of-band access to a router. It is used to set up the initial configuration of a router and to monitor it.

A rollover cable and an RJ-45 to DB-9 adapter are used to connect a PC to the console port. As you know from your previous studies, terminal emulation software is used to configure the router over the console connection.

Step 2: Which command is used to enter privileged EXEC mode?

enable

Step 3: Enter global configuration mode. Run the following command and write your observations.

Router#*configure terminal*

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#

we are now in global configuration mode where we can change the router's settings.

Task 3: Perform basic Configuration of Router R1.

Step 1: Configure the router name as R1.

Enter the command at the prompt and write your observations.

Router(config)#*hostname R1*

Router(config)#hostname R1
R1(config)

we give hostname to the router R1

Step 2: Disable DNS lookup.

Following command is used to disable DNS lookup.

R1(config)#*no ip domain-lookup*

Why would you want to disable DNS lookup in a lab environment?

mistype command causes a long delay while the router waits for a DNS response.

What would happen if you disabled DNS lookup in a production environment?

DNS resolution is needed for hostname-to-IP translation for management tasks like remote access and logging.

Step 3: Configure the EXEC mode password.

Configure the EXEC mode password using the following command. Use class for the password.

R1(config)#enable secret **class**

OR

R1(config)#enable password **class**

Write difference between setting password using enable secret and enable password?

1. enable secret class : password is stored encrypted in the config file

2. enable password class : password is stored in plain text

note : if both are set, enable secret takes priority

Step 4: Configure the console password on the router.

Run following commands to set console password on the router. Use cisco as the password. When you are finished, exit from line configuration mode. Write down your observations.

R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#

we set the password (cisco) to the line console 0, form now while connecting to router via console port the pass must be demanded

Step 5: Configure the password for the virtual terminal lines.

Run following commands to set telnet password on the router. Use cisco as the password. When you are finished, exit from line configuration mode.

R1(config)#line vty 0 4 5 user can access the router at a single time
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#

we set the password (cisco) to the line vty 0 4, form now while connecting to router remortly the pass must be demanded.

Step 6: Configure the FastEthernet0/0 interface.

Configure the FastEthernet0/0 interface with the IP address 192.168.1.1/24 by using following commands. Write down your observations

R1(config)#interface fastethernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

we interface the fe 0/0 wire , and assign ip addr , and set this connection to no shutdown the connection between router0 and switch is established (arrows get green)

Step 7: Configure the Serial0/0/0 interface.

Discuss the difference between DCE and DTE.

Data Terminal Equipment : Endpoint device that generates orreceives data

Data Communications Equipment : Intermediate device that facilitates communication

Why it is necessary to set one device as DCE and other as DTE on serial interface link?

a single clock source for synchronization, ensure proper hardware handshaking otherwise, communication cannot occur.

Configure the Serial0/0/0 interface with the IP address 192.168.2.1/24. Set the clock rate to 64000.

R1(config-if)#interface serial 0/0/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0

R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#exit
What is the significance of setting clock rate?

sets the transmission speed on the DCE device to synchronize data transfer between serial interfaces

Step 8: Use the following command to provide a description for this interface.
R1(config-if)#description Link to R2
What is the significance of description for the interface?

it provides the description to this connection and save in configration files

which helps in debuging

Step 9: Save the R1 configuration.
Save the R1 configuration using the following command and write your observations
R1#**copy running-configstartup-config**

it shows prompt :Destination filename [startup-config]? (Pressing Enter accepts the default)
file saved to NVRAM as the startup configuration that will load on next reboot

What is a shorter version of this command?

1. copy run start 2. write memory (wr)

Task 4: Perform Basic Configuration of Router R2.

Step 1: For R2, repeat Steps 1 through 6 from Task 3
Step 2: Configure the Serial 0/0/0 interface with the IP address 192.168.2.2/24.
Write commands for configuration.

```
R2(config)# interface s 0/0/0
R2(config-if)# ip address 192.168.2.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

Step 3: Configure the FastEthernet0/0 interface with the IP address 192.168.3.1/24. Write commands for configuration.

```
R2(config)# interface fe 0/0
R2(config-if)# ip address 192.168.3.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
```

Step 4: Save the R2 configuration.
Save the R2 configuration as done in Task 3 step 9.

Task 5: Configure IP Addressing on the Host PCs.

Step 1: Configure the host PC1.
Configure the host PC1 that is attached to R1 with an IP address of 192.168.1.10/24 and a default gateway of 192.168.1.1.
Step 2: Configure the host PC2.
Configure the host PC2 that is attached to R2 with an IP address of 192.168.3.10/24 and a default gateway of 192.168.3.1.
Explain the significance of gateway.

The default gateway is the exit point that allows a host to communicate with devices outside its own network

Task 6:Verify and Test the Configurations.

Step 1: Verify that routing tables have the following routes using the show ip route command.
The show ip route command and output will be thoroughly explored in upcoming labs. For now, you are interested in seeing that both R1 and R2 have two routes. Both routes are designated with a C. These are the directly connected networks that were activated when you configured the interfaces on each router. If you do not see two routes for each router as shown in the following output, proceed to Step 2.
Run the following command on R1 and R2 and write your observations.
R1#show ip route

displays the router's routing table, it shows the connected networklinks directly to the router.

```
C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial2/0
```

Step 2: Verify interface configurations.
Another common problem is router interfaces that are not configured correctly or not activated. Use the following command to quickly verify the configuration of each router's interfaces. Write your observations
R1#show ip interface brief

summary of all interfaces, showing their IP addresses and whether each interface is operationally up or down at both Layer 1 (Status) and Layer 2 (Protocol)

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.1.1	YES	manual	up	up
Serial2/0	192.168.2.1	YES	manual	up	up

Step 3: Test connectivity using ping.
The ping command is a useful tool for troubleshooting Layers 1 through 3 of the OSI model and diagnosing basic network connectivity. This operation can be performed at either the user or privileged EXEC modes. Using ping sends an Internet Control Message Protocol (ICMP) packet to the specified device and then waits for a reply. Pings can be sent from a router or a host PC. Use the ping command to test connectivity between the R1 router and PC1. Write your observation.
R1#ping 192.168.1.10

```
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
```

the connection is established between R1 router and PC1

Test connectivity by pinging from each host to the default gateway that has been configured for that host.
From the host attached to R1, is it possible to ping the default gateway? yes
From the host attached to R2, is it possible to ping the default gateway? yes

Task 7: Reflection

Step 1: Attempt to ping from the host connected to R1 to the host connected to R2.
This ping should be unsuccessful.
Step 2: Attempt to ping from the host connected to R1 to router R2.
This ping should be unsuccessful.
Step 3: Attempt to ping from the host connected to R2 to router R1.
This ping should be unsuccessful.
What is missing from the network that is preventing communication between these devices?

there is no routing table to forward requests in WAN thats why the pc1 isn't ping pc2

Task 8: Create a startup configuration file

Router configurations can be captured to a text (.txt) file and saved for later use. The configuration can be copied back to the router so that the commands do not have to be entered one at a time.
Step 1: View the running configuration of the router using following command.
R1#***show running-config***

```
Building configuration...
Current configuration : 956 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R0

enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
ip cef
no ipv6 cef
--More--
```

Step 2: Copy the command output.
Select the command output. From the HyperTerminal Edit menu, choose the copy command.
Step 3: Paste output in Notepad.
Open Notepad. Notepad is typically found on the Start menu under Programs > Accessories. From the Notepad Edit menu, click Paste.
Step 4: Edit commands.
Some commands will have to be edited or added before the startup script can be applied to a router. Some of these changes are:
Adding a no shutdown command to FastEthernet and serial interfaces that are being used.
Replacing the encrypted text in the enable secret command with the appropriate password.
Removing the mac-address command from the interfaces.
Removing the ip classless command.

Step 5: Save the open file in Notepad to start.txt.

Task 9: Load the start.txt File onto the R1 Router

Step 1: Erase the current startup configuration of R1.

Use following command and confirm the objective when prompted, and answer no if asked to save changes.

R1#***erase startup-config***

Step 2: When the prompt returns, issue the following command.

R1#***reload***

Confirm the objective when prompted. After the router finishes the boot process, choose not to use the AutoInstall facility.

Step 3: View the running and startup configuration of the router and write your observations.

_____

the start-config isn't present , running-config is present.

_____

Step 4: Enter global configuration mode.

Router#***configure terminal***

Step 5: Write the commands to verify the running configuration.

_____

show running-config

_____

Step 6: Write the commands to save running configurations

_____

copy running-config startup-config

_____

Rubric for Lab Assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed assigned tasks without any help from the instructor and showed the results appropriately.	4	
Good	The student completed assigned tasks with minimal help from the instructor and showed the results appropriately.	3	
Average	The student could not complete all assigned tasks and showed partial results.	2	
Worst	The student did not complete assigned tasks.	1	

Instructor Signature: _____ Date: _____